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FEDERAL UNIVERSITY OF LAFIA, FACULTY OF SCIENCE, DEPARTMENT OF PHYSICS

2ND SEMESTER 2023/24 SESSION TEST

Course Title: General Physics IV (Vibration, Waves and Optics) (PHY104)

Instruction: Answer all questions. **Time allowed:** 1 hr

1. Which of the following statements is true about Simple Harmonic Motion (SHM)? **(a)** The motion is always linear **(b)** The motion is always linear **(c)** The motion is periodic and oscillatory **(d)** The motion does not involve any restoring force

2. The motion of a particle executing SHM, is given by $x = 0.01 \sin 100\pi(t + 0.05)$. Where x is in meter and time t is in seconds. The time period is **(a)** 0.01 sec **(b)** 0.02 sec **(c)** 0.1 sec **(d)** 0.2 sec

3. The time period of a mass-spring system in SHM depends on: **(a)** The amplitude of motion **(b)** The initial displacement from equilibrium position **(c)** The mass of the object only **(d)** The mass of the object and the spring constant

4. For an object in SHM, the maximum displacement from the equilibrium position is called: **(a)** Amplitude **(b)** Frequency **(c)** Wavelength **(d)** Period

5. A body is executing simple harmonic motion with an angular frequency 2 rad/sec. The velocity of the body at 20 mm displacement when the amplitude of motion is 60 mm is **(a)** 40 mm/sec **(b)** 60 mm/sec **(c)** 113 mm/sec **(d)** 120 mm/sec

6. When the displacement of an object in SHM is at its maximum, its velocity is: **(a)** Mid value **(b)** At its minimum **(c)** At its maximum **(d)** Not enough information to determine

7. Kinetic energy of SHM is ________ at mean position **(a)** zero **(b)** minimum **(c)** constant **(d)** maximum

8. ________ is maximum and equal to total energy at extreme positions in SHM **(a)** kinetic energy **(b)** mechanical energy **(c)** potential energy **(d)** kinetic and potential energy

9. A particle is executing SHM if its amplitude is 2 m and periodic time 2 $seconds$. Then the maximum velocity of the particle will be **(a)** 2π **(b)** 6π **(c)** 4π **(d)** π

10. A body is moving in a room with a velocity of 20 m/s perpendicular to the two walls separated by 5 meters. There is no friction and the collision with the walls are elastic. The motion of the body is **(a)** Not periodic **(b)** Periodic but not simple harmonic **(c)** Periodic and simple harmonic **(d)** Periodic with variable time period

11. what happens when a red apple is illuminated by a green flashlight in a dark room (a) the apple will reflect all of the colors in white light except red and thus it appears red (b) the apple will absorb all of the colors in white light except red and thus it appears red (c) the apple will not absorb any colors (d) the apple will only absorb red color

12. Which of the following are interference pattern? a) destructive interference b) constructive interference c) white light d) all of the above

13. If a yellow banana absorbs all other colors in white light, what color will reflect? - Yellow

14. What would be the angle of incidence and angle of refraction for a beam of light not to deviate in glass? a) 90 degrees both b) exactly 180 and 90 c) zero and 90 degree d) zero both

15. According to Huygens's principle, the ether medium pervading entire universe is a) less elastic and denser b) highly elastic and less dense c) not elastic d) much heavier and dense

16. Huygens's concept of secondary waves (a) allows us to find the focal length of the thick lens b) it is a geometrical method to find the wavelength c) used to determine a velocity of light d) it is used to explain polarization

17. A convex lens with a focal length of 15cm is placed from lighted object. At what distance will an image be formed? a) 8 cm b) 3 c) 7.5 cm d) 6.5 cm

18. Where should one place a lighted object so that the final image is 4m away from a lens having a focal length of 20cm? a) 20 cm b) 22cm c) 9 cm d) 21.05 cm

19. Will the image in question 2 above be enlarged or reduced? a) the image will be smaller b) the image will be enlarged c) the image will be doubled a) the image will be reduced

20. We placed an object 20 cm from a convex lens with a focal length of 25cm. where will the image be formed? a) 10 cm b) 1 cm -10 cm d) -100 cm

21. -------- is the resultant wave due to the interference of two (2) sinusoidal transverse waves, it is also a sinusoidal transverse wave with an amplitude and an oscillating term. (a) $2 \sin ½(\alpha + \beta) \cos ½(\alpha - \beta)$ (b) $y_R(x, t) = [2A \cos ½\phi] \sin(kx - \omega t + ½\phi)$ (c) $\cos ½(kx - \omega t - kx + \omega t - \phi)$ (d) $A \sin (kx - \omega t)$

22. The combination of separate waves in the same region of space to produce a resultant wave is called ---------- (a) transverse wave (b) sinusoidal (c) interference (d) amplitude

23. Two waves are described by: $y_1 = 0.30 \sin[\pi(5x - 200t)]$, $y_2 = 0.30 \sin\left[\pi(5x - 200t) + \frac{\pi}{2}\right]$. Where y_1, y_2 *and* x are in meters and t is in seconds. When these two waves are combined, a traveling wave is produced. What are the is Amplitude of the traveling wave (a) 0. 3424m (b) 0.4243m (c) 0.243m (d) 0.423m.

Use question (23) to answer (24) and (25)

24. The Wave speed of the traveling wave is (a) 401 m/s (b) 590 m/s (c) 10 m/s (d) 40 m/s

25, Wavelength of that traveling wave is (a) $0.4m$ (b) $0.3m$ (c) $0.1m$ (d) $0.23m$

26. The bending of waves around corners of an obstacle or aperture is called .... (a) diffraction (b) interference (c) transverse (d) sinusoidal

27. Polarization is a proof that light is a travel in a straight line, and it occur only in .... (a) transverse waves (b)longitudinal waves (c) linear wave (d) super positional wave

28. The constant frame of reference of the velocity of light in a vacuum is a universal with the value of -------------- approximately (a) $3.0 \times 10^5\ m/s$ (b) $2.0 \times 10^{10}\ m/s$ (c) $3.0 \times 10^2\ m/s$ (d) $3.0 \times 10^8\ m/s$

29. Velocity of light depends upon the medium through which it travels and is denoted by (a) $c = f\lambda$ (b) c= f/λ (c) $c = \lambda/f$ (d) $c = 2f\lambda$

30, The range of sound or audio frequency for Humans lies between (a) 15 – 20,000 Hz (b) 15 – 34 Hz (c) 12 – 23,000 hz (d) 14 – 140hz

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Instruction: Answer all questions   Time allowed: 1 hr

1. Velocity of sound wave in solids is mathematically represented as:
(a) $V=\sqrt{\frac{E}{\rho}}$ (b) $v\sqrt{\frac{T}{m}}$ (c) $V=U+ut$ (d) $V=\sqrt{\frac{B}{\rho}}$

2. Velocity of sound wave in liquids is mathematically represented as:

(a) $V=\sqrt{\frac{E}{\rho}}$ (b) $v\sqrt{\frac{T}{m}}$ (c) $V=U+ut$ (d) $V=\sqrt{\frac{B}{\rho}}$

3. What is the speed of sound in an aluminium bar whose modulus is $7.0\times10^{10}\ N/m^2$ for aluminium and its density is $2.70\times10^3\ kg/m^3$. (a) $5292\ m/s$ (b) $4092\ m/s$ (c) $5092\ m/s$ (d) $5002\ m/s$

4. What is the bulk modulus of a distilled water, if $32.0\ g$ of distilled water occupies 22.4 *litres* and the speed of the sound in the distilled water is $1480\ m/s$. (a) $3.13\times10^9\ N/m^2$ (b) $3.\times10^3\ N/m^2$ (c) $2.16\ N/m^2$ (d) $13\times10^3\ N/m^2$

5. When tuning forks of frequency $206\ Hz$ and $194\ Hz$ are sounded simultaneously. What is the frequency of the resultant beat. (a) 24 hz(b) 13 hz(c) 12 hz (d) 16 hz

6. The reflection of sound waves when they meet a hard surface/object is called -------- (a) pitch (b) echo (c) noise (d) resonance

7. The following are important of echo except ---------- (a) They are used to determine the distance of an ocean/sea. (b) They are used to detect cracks or faults in metal casting. (c) They are also used in oil/petroleum prospective. (d) They are used in military operations for the detection of enemies in sub-marines at sea level.

8. In a resonance tube experiment, a tuning fork of frequency 256 Hz gave resonance when the water level was 32.5 cm below the open end of the tube. If the next position of resonance was 100 cm, what is the speed of sound in air? (a) $7.8\ m/s$ (b) $5.6\ m/s$ (c) $345.6\ m/s$ (d) $45.6\ m/s$

9. What is the frequency of the first overtone of a closed pipe of length20 cm, if the speed of sound in air is $340\ m/s$. (a) 78 hz (b) 1275 hz (c) 7899hz (d) 456 hz

10. The unit of a sound intensity is ------------ (a) Watts/m^2 (b) Watts (c) Joules.Watts/m^2 (d) Watts/m^{2}h

11. A converging lens has a focal length of 10 cm. if a light is located 30 cm from the lens, what type of image will be formed? a) smaller, inverted and a real image will be formed b) larger, inverted and a real image will be formed c) a virtual and erect image will be formed d) smaller, upright and a real image will be formed

12. Where will the image in question 24 be? a) image will be further away from 2F b) image will be on 2F c) same size and at the center d) image will be further away from F

13. A person who want to be able to see an ant holds it 4cm from a 6 cm focal length magnifying glass. Find the magnification of the lens. a) -2 b)+3 c) -3 d) +4

14. How does positive magnification affect image formed? a) image becomes virtual and erect b) image becomes real and inverted c) image becomes diminished d) image becomes magnified

15. A concave lens of focal length 15cm forms an image 10 cm from the lens. How far is the object placed from the lens? a) 20 cm b) 25 cm c) 30 cm d) 35 cm

1. Which of the following is a necessary condition for total internal reflection?

a) The angle of incidence in the denser medium must be greater than the critical angle for the two media

b) The angle of incidence in the rarer medium must be greater than the critical angle for the two media

c) The angle of incidence in the denser medium must be lesser than the critical angle for the two media

d) The angle of reflection in the denser medium must be greater than the critical angle for the two media

2. Why is the sequence of colors in the secondary rainbow reverse of that in the primary rainbow?

a) Refraction of light b) Two internal reflection c) Reflection of light d) Dispersion

3. 'X' is an optical illusion observed in deserts or over hot extended surfaces like a coal-tarred road, due to which a traveler sees a shimmering pond of water some distance ahead of him and in which the surrounding objects like trees appear inverted. Identify X.

a) reflection illusion b) Mirage c) Optical illusion d) Total internal reflection

4. Identify the principle behind the sparkling of diamonds.

a) Total internal reflection b) Refraction c) Reflection d) Optical cavity

5. The ratio of the velocity of light in a medium to the velocity of light in a vacuum is 4/5. If the ray of light is emerging from this medium into the air, the critical angle for this interface of medium and air will be

a) 90^0 b) 37^0 c) 53^0 d) 45^0

6. In total internal reflection when the angle of incidence is equal to the critical angle for the pair of media in contact, what will be the angle of incidence?

a) 90 degrees b) 180 degrees c) 0 degrees d) equal to the angle of incidence

7. If A, B and C are the critical angles of glass- air interface for red, violet and yellow color, then

a) C greater then B greater A b) A greater then B greater than C c) A equal to B equal to C d) A great er than C greater than B

8. A ray of light is seen to be reflected from a plane mirror at an angle of

30°with the mirror. The angle of incidence must be

a) 30° b) 120° c) 90° d) 60°

9. An object is placed at a distance of 10cm from a convex mirror of focal length 15cm find the posit ion and nature of the image

a) v = -6cm, real & inverted b) v = -3cm, real & inverted c) v = +6cm, virtual & erect d) v = +15cm, virt ual & erect

10. An object 8cm in length is placed at a distance of 40cm in front of a concave mirror of radius of curvature 30cm. Find the position of the image and its size

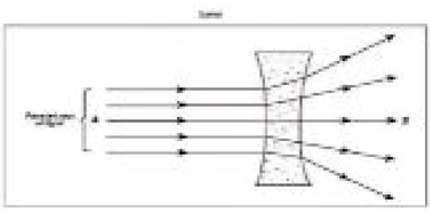
a) v = -24cm, h = -4.8cm b) v = 24cm, h = 4.8cm c) v = -32cm, h = -5.2cm d) v -32cm, h = 5.2cm

12. When a ray of light moves from the air to water it bends due to a change in

a) Velocity b) Direction c) Speed d) refraction

13. When light passes through this object, it is refracted off of one wall and reflected to another, the light is refracted again and as it leaves the object, it forms a rainbow. What is this object?

a) Mirror b) Magnifying glass c) Prism d) Window



14. What type of image does this lens form?

a) Virtual image b) Real Image c) No Image d) inverted image

15. If I were to stand in front of a flat mirror 5m away how many would I appear on the flat mirror

a) 10m away b) 5m away c) 0m away d) 1m/s away

16. A 25.0 m object is placed 30.0 m from a convex lens, which has a focal length of 5.0 m. What is t

he distance of the image?

a) 6 m b) 7 m c) 8 m d) 9 m

17. What is the index of refraction of a refractive liquid if the angle of incidence in the air is 35 degre es and the angle of refraction is 14 degrees?

a) 1.33 b) 2.21 c) 2.37 d) 1.72

18. Real images are :

a) Always inverted b) Magnified or Diminished c) can be obtained on screen d) all the above

19. A dentist uses a small dental mirror to help magnify teeth in your mouth. What type of mirror is this?

a) a convex mirror b) a concave mirror c) a convex lens d) a concave lens

20. A student is 2.50m away from a convex lens while her image is 1.80m from the lens, what is the focal length?

a) 1.05m b) 2.50m c) 1.39m d) 0.72m

21. An object is placed at a distance of 27.0 cm away from a thin convex lens with a focal length of 9.00 cm. How far from the lens is the image located and what type of image is formed?

a) 13.5 cm and real b) 3.00 cm and real c) 0.33 cm and virtual d) 0.15 cm and virtual

22. 10. An object is positioned between two plane mirrors inclined at straight line angle to each othe r. The object is 1 unit distance from each mirror, the number of images formed is

a) 1 b) 2 c) 3 d) 4

23. An object 1cm high placed on the axis 15cm from a converging lens forms an image 30cm from the lens. The size of the image is

a) 0.5cm b) 1.5cm c) 2.5m d) 2cm

24. By what amount in terms of angular distance does a reflected ray shift when a plane mirror is sh ifted through an angle of 2°.

a) 2° b) 4° c) 6° d) equal to the angle rotated by the plane mirror surface

25. Magnification produced by a convex mirror is always

a) <1 b) >1 c) > 1 or = 1 d) <1 or > 1

26. What is the best analogy for what happens to light in a chromatic aberration?

a) A telescope focusing on a faraway object. b) A prism dividing light into waves of color. c) A magnifying glass that focuses light d) an unfocused floodlight that casts light everywhere.

27. The critical angle for water when the refracted angle is 90^0 and refractive index for water and air are 1.33 and 1 respectively, is?

a) 48.8^0 b) 49.5^0 c) 50^0 d) 51^0

28. When a ray of light enters from a denser medium to a rare medium, it bends

a) towards normal b) away from normal c) perpendicular to normal d) parallel to normal

32. Which of the phenomenon takes place inside of an optical fibre

a) reflection b) dispersion c) total internal reflection d) scattering

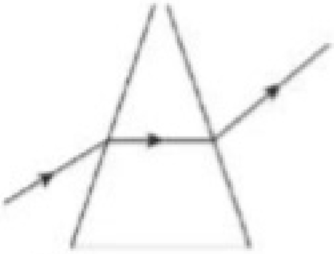
29. The speed of light in medium M_1 and M_2 are 1.5×10^8 m/s and 2.0×10^8 m/s respectively. A ray of light enters from at an angle of incidence *i*. If the ray suffers total internal reflection, the value of *i* is

a) equal to Sin^{-1} (2/3) b) equal to or less than Sin-1 (3/5) c) equal to or greater than Sin^{-1} (3/4) d) less than Sin^{-1} (2/3)

30. What happens to the refractive index on heating the liquid

a) increases b) decreases c) remain unchanged d) increases or decreases depending on the rate of heating.

Which imagine shows deviation of light in glass prism?



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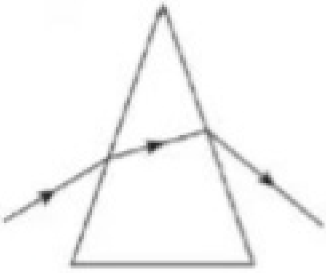

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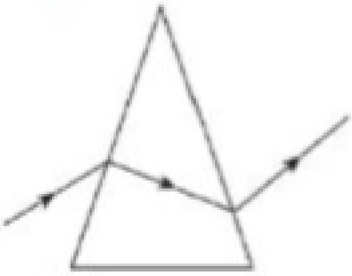

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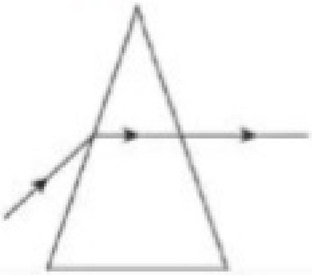

32. An object is placed between the focus and pole of a double convex lens. The image is

a) virtual, erect and magnified b) virtual, erect and diminished c) real, inverted and magnified d) real, erect diminished

33. An image formed by a single diverging lens

a) is upside down b) can be projected on the wall c) is virtual d) is larger than the object

34. f = R/2 is valid for

a) convex mirrors only b) concave mirrors only c) plane mirrors d) convex mirrors and concave mirrors

35. An endoscope uses glass fibers to enable doctors to see inside a person. Rays of light travel along the glass fibers by...

a) optical transmission b) ray penetration c) total internal reflection d) ray interaction by reflection

36. The nature of image formed in the rear mirror of a car is

a) inverted image b) Erect image c) real image d) None of the above

37. an object is places at a distance of [illegible] m from the principal focus of focal length fm. What will be the ratio of height of image to height of object?

a) [illegible] b) [illegible] c) [illegible] d) [illegible]

38. To get an erected and 16 times magnified image of object, object is needed to be placed at what distance in front of a concave mirror?

a) 25fcm b) [illegible] c) 16fcm d) 17fcm

39. Point object at a distance of 4f from the pole of a convex mirror whose image is formed at a distance of 4f behind the mirror, how many times greater will the image be?

a) 2 b) 4 c) 6 d) 8

40 a point object is placed at a distance of 3f from the pole of a concave mirror then moved away by a distance of 2f. What would be the ratio of the image formed?

a) 25 b) [illegible] c) [illegible] d) [illegible]

40. what would be the refractive index of a medium when the incident ray in a medium is refracted and travelling along the surface separating the two medium?

b) [illegible] b) [illegible] c) [illegible] d) [illegible]

EXERCISE

- The angle of a prism is 60^0. what is the angle of incidence for a minimum deviation? The refractive index of the material of the prism is $\sqrt{2}$.
- The refractive index of the material of a triangular glass prism is $\sqrt{3}$. What is the angle of incidence and the angle of minimum deviation of a ray of light incident upon one face of the prism

EXERCISE

- What is the position, nature and size of the image of an object of size 9cm placed in front of a diverging mirror of focal length f at a distance of 3f/2?
- To get erected and 16 times magnified image of an object, the object should be placed at what distance from a concave mirror?